

5.3.5.2 Concrete pipes

General

Unreinforced and reinforced concrete pipes and fittings with flexible joints shall comply with TS 821 EN 1916. The pipes shall be reinforced to stand the load during installation as well as in the permanent installation. Concrete pipes shall be sealed with standard rubber ring gaskets. Prior to use the jointing rings shall be stores in a cool place, protected from direct sunlight and frost.

Pipes used for wastewater pipelines shall be steam cured pipes with integrated gasket type.

Pipes used for stormwater pipelines shall be steam cured with separate gasket type.

Pipes shall be subject to rejection on account of failure to meet any of the testing requirements. Pipe sections also may by rejected if found to contain cracks, damaged ends and other defects due to faulty manufacture, handling, transporting or placing.

For pipe diameters of 200 mm to 400 mm, connections to the wastewater pipe can only be performed with single branch fittings produced in a factory. For pipe diameters of 500 mm and greater, saddles may be used but only saddles using direct drilling in the concrete pipe.

Pipe diameter, mm (Pipe with bell and integrated gasket)	Minimum compressive strength, kN/m (Force applied on crest)
200 mm concrete	45
300 mm. concrete	48
400 mm. concrete	54
500 mm. concrete	60
600 mm. concrete/reinforced concrete	65
800 mm. reinforced concrete	72
1000 mm. reinforced concrete	80
1200-3500 mm. reinforced concrete	According to TS-821 EN 1916

Pipe Length, joints and reinforcement

Standard pipe lengths shall not be shorter than the values specified below (excluding special and short pieces).

Diameter, mm	Minimum pipe length, m
600 and smaller	1.25
700 - 1000 and larger	2.00

Every concrete and reinforced concrete pipe joint shall be designed to withstand a load applied on joint, load being equal to the gasket pressure plus 60,000N per every 1 m of internal diameter of the pipe.

Spiral reinforcement shall be of totally circular type. Ellipsoid or partial circular reinforcement shall not be accepted. Longitudinal reinforcement shall not be less than 0.2% of the pipe cross sectional area. Longitudinal reinforcement shall be placed around the pipe in a uniform spacing.